



Unboxing the Black Box: A Survey on Mechanistic Interpretability for Algorithmic Understanding of Neural Networks

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Abstract

The black box nature of deep neural networks poses a significant challenge for the deployment of transparent and trustworthy artificial intelligence (AI) systems. With the growing presence of AI in society, it becomes increasingly important to develop methods that can explain and interpret the decisions made by these systems. To address this, mechanistic interpretability (MI) emerged as a promising and distinctive research program within the broader field of explainable artificial intelligence (XAI). MI studies the inner computations of neural networks and translates them into human-understandable algorithms. It encompasses reverse-engineering techniques aimed at uncovering the computational algorithms implemented by neural networks. In this paper, we present a comprehensive survey of mechanistic interpretability, synthesizing a rapidly growing and fragmented body of work into a single, structured reference. We propose a unified taxonomy of MI approaches and provide a detailed analysis of key techniques, illustrated with concrete examples and pseudo-code. We contextualize MI within the broader interpretability landscape, comparing its goals, methods, and insights to other strands of XAI. Additionally, we trace the development of MI as a research area, highlighting its conceptual roots and the accelerating pace of recent work. We argue that MI has the potential to support a more scientific understanding of machine learning systems – treating models not only as tools for solving tasks, but also as systems to be studied and understood. We intend this survey to serve as an entry point for new researchers to the field of mechanistic interpretability.

Keywords Explainable AI · Mechanistic interpretability · Survey

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1 Introduction

Artificial intelligence (AI) is increasingly being deployed in a wide range of tasks, from everyday applications like recommendation systems to high-risk domains such as biometric recognition, autonomous vehicles, and medical diagnosis (Ali et al., 2023). The rise of transformer-based models, particularly in natural language processing (NLP), has further accelerated this adoption, enabling breakthroughs in text generation, translation, and image understanding (Lin et al., 2022). However, the size, complexity, and opacity of these models are growing just as quickly, outpacing researchers' ability to understand their internal workings. This combination of growing influence and persistent opacity has led to a heightened demand for AI models that are both faithful and explainable. The validation of AI-generated decisions is especially critical in high-risk areas, such as law or medicine (Barredo Arrieta et al., 2020; Srinivasu et al., 2022). Consequently, Explainable AI (XAI) emerged in response to the companies' and researchers' demands to interpret, explain, and validate neural networks to make AI systems trustworthy.

XAI encompasses all methods and approaches to uncover the reasoning and behavior of artificial intelligence systems (Ali et al., 2023; Arsenault et al., 2025; Theissler et al., 2022). Thus, it is important to establish an understanding of common terms used in the XAI literature, despite the lack of universally accepted definitions. In general, *explainability* refers to understanding a system's outputs (Hamida et al., 2024), answering the *why?* question; whereas *interpretability* refers to understanding the system's internal workings (Das, 2020; Räuker et al., 2023), answering the *how?* question. However, this distinction is not universally agreed upon. Lipton (2018) argues that interpretability bundles several distinct properties rather than a single, well-defined one, while Rudin (2019) draws a sharper distinction between *interpretable* models – which are transparent by design – and black-box models that have merely been *explained* after the fact, cautioning against treating the two as equivalent. Throughout this survey, we adopt the above definitions for consistency, while acknowledging this terminological plurality.

Interpretable artificial intelligence refers to AI models that are intrinsically transparent in their operation, eliminating the need for post-hoc analysis. Consequently, these models enable the identification and tracking of causal relationships (Schwalbe & Finzel, 2024). *Trustworthy* AI is mainly associated with AI systems designed to inspire trust by prioritizing safety, fairness, and transparency. The use of XAI is a viable approach to building trustworthy AI models (Kowald et al., 2024; Markus et al., 2021). Table 1 summarizes the key aspects of XAI.

The growing demand for transparency in AI systems has led to the development of a vast number of Explainable AI methods, each designed to address different aspects of interpretability across various models and applications. Researchers have proposed differ-

Table 1 Different aspects of XAI

Aspect	Explainable AI	Interpretable AI	Trustworthy AI
Focus	Post-hoc explanations of models' decisions	Transparent models by design	Safe and ethical AI systems
Goals	Explaining the model's decision	Understanding model functioning	Building trust in AI systems
Examples	Feature importance	Decision trees	Bias-free hiring system

ent taxonomies to organize this diversity (Saranya & Subhashini, 2023; Ali et al., 2023; Barredo Arrieta et al., 2020; Schwalbe & Finzel, 2024). Two key distinctions recur across these taxonomies: *scope* (local explanations for individual predictions versus global insights into overall model behavior) and *stage of application* (whether explanations are integrated into the model or applied post-hoc). Within post-hoc approaches, a further distinction separates model-specific methods, tailored to a particular model, from model-agnostic methods, which operate independently of the underlying model (Speith, 2022). Other distinctions may include the format of the explanation produced, the type of input data (e.g., tabular, image, text), or the underlying problem type (e.g., classification, regression). Figure 1 presents the key aspects of this taxonomy of XAI methods.

One notable approach that emerged over the past few years is Mechanistic Interpretability (MI), a cognition-based set of reverse-engineering techniques aimed at uncovering the computational algorithms of neural networks (Olah, 2022; Saphra & Wiegrefe, 2024). Mechanistic interpretability focuses on the internal structures of AI models – neurons, layers, attention heads, and circuits – and their causal roles in computation. Unlike explainability methods, such as feature attribution (e.g., LIME Ribeiro et al. 2016 or SHAP Lundberg & Lee 2017), MI seeks a deeper understanding of models, drawing inspiration from neuroscience and systems biology. According to the proposed taxonomy, MI is typically post-hoc and model-specific, similar to feature-attribution methods. However, MI differs in its primary objective: attribution methods are largely local and correlational, identifying which inputs influenced an output, whereas MI pursues global, causal accounts of the algorithm a model implements internally. This shift from "what mattered" to "how it works" is the central distinguishing contribution of MI within the broader XAI landscape. Rather than treating AI systems as a black boxes, MI researchers emphasize the importance of inner interpretability based on the premise that internal components of neural networks adopt specific roles after training (Manning et al., 2020; Saphra, 2021). Identifying such components provides deeper insight into model behavior and a scientific understanding of AI. To uncover these roles, researchers employ a range of techniques rooted in pattern recognition, as any non-random structure within a neural network can serve as a functional component (Kästner & Crook, 2024). These techniques include methods for analyzing the representations learned by models, assessing the causal impact of specific components, and visualizing patterns within the network's structure and activations.

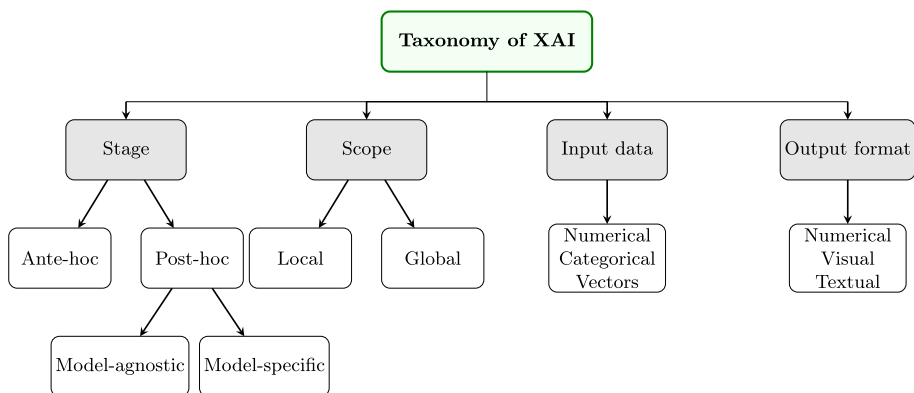


Fig. 1 Brief taxonomy of XAI methods

Mechanistic interpretability provides a systematic characterization of AI systems as opposed to ad hoc methods of XAI, which provide explainability only in specific contexts (Kästner & Crook, 2024). Its primary goal is to improve AI safety and build trust by providing thorough, human-interpretable explanations of complex AI models. This is achieved by mapping low-level components, such as circuits or neurons, to elements of human-understandable algorithms and building upward to develop a holistic system-level understanding (Olah, 2022). As research in this area progresses, current efforts are increasingly focused on refining methodologies and scaling analyses to more complex models. The increasing interest in mechanistic interpretability is further evidenced by the growing number of publications, particularly review articles. Kästner and Crook (2024), Grzankowski (2024), and Rabiza (2024) situate mechanistic interpretability in a philosophical context, rather than surveying its technical methods. Davies and Khakzar (2024) seek parallels between MI and cognitive science. Bereska and Gavves (2024) present a comprehensive overview of advancements in MI, along with a taxonomy of concepts, but frame their contribution primarily around AI safety rather than situating MI within the wider XAI landscape. Rai et al. (2025) focus on employing MI in language models, and Saphra and Wiegrefe (2024) present the divide between MI and NLP interpretability communities. Lin et al. (2025) survey MI in multimodal models and identify open research gaps in that subdomain. Sharkey et al. (2025) focus on open challenges in the field and present possible future directions, without providing a taxonomy of existing techniques.

Taken together, existing surveys are either philosophically oriented, safety-focused, or scoped to a particular modality or community, and none systematically position MI within the broader XAI field while also providing a comprehensive, technique-level taxonomy. Our survey addresses this gap: we survey the current status of mechanistic interpretability, position MI within the broader XAI community, and propose a clear definition along with a taxonomy of techniques that spans the modalities and tasks discussed across the works above. We aim to organize the rapidly growing body of mechanistic interpretability research into a coherent conceptual and methodological framework. Despite its growing importance, the field lacks a consolidated reference that systematically covers its foundations, methods, and open challenges. This survey aims to fill this gap by providing a comprehensive overview of mechanistic interpretability. Specifically, we pursue the following objectives:

- Situating mechanistic interpretability within the broader context of mechanistic explanations and explainable AI (Sect. 2).
- Providing a clear definition and taxonomy of mechanistic interpretability (Sect. 2).
- Providing a comprehensive literature review on mechanistic interpretability (Sect. 3).
- Presenting the key techniques used in the field, with pseudo-code and literature examples (Sect. 4).
- Outlining major challenges and opportunities in MI (Sect. 5).

The paper is structured as follows: Sect. 2 describes the philosophical background of MI and places it in the broader XAI community. It follows with a brief introduction to the core concepts in MI. Section 3 presents a literature review. Section 4 describes the different approaches used in MI and presents notable advancements. Section 5 addresses the challenges and opportunities found in MI. Section 6 outlines future goals in MI and concludes the article.

2 What is Mechanistic Interpretability?

The mechanistic approach to understanding neural networks is a relatively new development within the field of AI interpretability. As this approach gains traction, it is important to situate it within the broader XAI community. In this section, we first outline the scientific and historical context of MI, followed by the presentation of a definition and taxonomy.

2.1 Origins of Mechanistic Interpretability

Philosophical background. The study of mechanisms and mechanistic explanations is rooted in science and philosophy and has been a topic of research since the 20th century (Craver et al., 2024). Bechtel and Abrahamsen (2005) define a mechanism as a *structure performing a function in virtue of its components parts, component operations, and their organization*. In other words, a mechanism is a organized structure that causes some phenomena through interactions between parts, and can be represented as a directed graph of components involved in the process. By studying the phenomenon, researchers aim to uncover the underlying mechanism and explain its functioning in terms of its causal components. This investigative process culminates in the formulation of a mechanistic explanation (Machamer et al., 2000). In fields like medicine (Courcelles et al., 2022; Metzcar et al., 2024), physics, and neuroscience (Bassett & Sporns, 2017), mechanistic explanations play a critical role in dissecting complex systems, revealing their constituent parts, and establishing causal relationships that elucidate how these components work together to produce emergent behavior (Miller, 2019).

Mechanistic interpretability parallels the way scientists study physical systems and aligns AI research with the foundational principles of explanatory science. It conceptualizes artificial neural networks as complex systems composed of multiple mechanisms, establishing an analogy between AI and the human brain (Davies & Khakzar, 2024; Liu et al., 2023; Fernando & Guitchounts, 2025; Mineault et al., 2024). This approach is further supported by the previous applications of mechanistic explanations in neuroscience and cognitive sciences (Craver, 2007; Craver et al., 2024; Kostic & Halfman, 2023; Khambhati et al., 2018). By bridging insights from neuroscience and AI, mechanistic interpretability offers a new framework for understanding artificial neural networks.

XAI background. Mechanistic interpretability first emerged in the context of interpreting vision models (Olah et al., 2020). With most of its early research being published on blogs and forums (see, e.g., LessWrong), the field developed separately from other interpretability methods. At the same time, due to the rise of advanced transformer models, the broader XAI community focused its attention on the explanations of large language models. As a result, an NLP interpretability community emerged, which at the time was mostly unaware of the mechanistic research. Concurrently, many MI researchers have turned their attention to interpreting transformers (Elhage et al., 2021). Thus, two distinct communities emerged, united by the same goal: AI safety (Saphra & Wiegrefe, 2024).

When MI researchers engaged with academia, a clash with the NLP interpretability community emerged. Initially, many NLP researchers criticized MI research for "rediscovering" existing interpretability methods that had already been adopted and refined within the NLP community. This tension was fueled by differing terminologies and methodological approaches, leading to debates over the novelty and relevance of MI contributions. NLP

researchers argued that MI often overlooked the rich history and established practices of NLP interpretability, while MI proponents contended that their focus on detailed, mechanistic explanations provided deeper insights into model behavior. Despite these conflicts, the clash ultimately spurred a productive dialogue, encouraging both communities to re-evaluate and integrate their methods, fostering a more comprehensive understanding of AI interpretability.

Although both mechanistic and NLP interpretability communities work towards similar goals, the term "mechanistic" has not been widely adopted in academia. Interpretability methods may or may not be described as mechanistic, leading to a blurred line between taxonomies and categorizations of approaches – for instance probing, which was present long before MI, has been adopted by MI researchers and is now often classified as a mechanistic method (Alain & Bengio, 2017; Gurnee et al., 2023). This overlap has led to a rich, albeit complex, landscape of interpretability research, where methods and terminologies often overlap and evolve.

2.2 Defining Mechanistic Interpretability

Following the discussion in the previous section, we propose a new definition for mechanistic interpretability: one that integrates MI with the overall **inner interpretability** research, but distinguishes it by focusing on the computational aspects of neural networks.

Definition. *Mechanistic interpretability is the process of studying the inner computations of neural networks and translating them into human-understandable algorithms.*

This definition supports the core idea of MI, which is its algorithmic approach to neural networks. Olah (2022) compares MI to reverse-engineering a compiled computer program. Similarly to variables interacting in programs to produce a desired output, neural networks are composed of neurons, interacting through their connections via weights. Based on the premise that neurons can be understood, and therefore the connections between them, MI aims to deconstruct neural networks into interpretable subnetworks. Although a neural network as a whole is a complex structure, it can be broken down into small, meaningful components.

The fundamental unit of neural network representations is a **feature**. Features refer to any interpretable properties or characteristics encoded in neurons. Unlike traditional machine learning, where features correspond directly to input data attributes (Bishop, 2006), MI treats them as abstractions capturing internal model properties. For instance, a feature in a vision model could be a curve detector, which responds to images containing a curve (Olah et al., 2020). Formally, they are defined as directions in the network's activation space (Elhage et al., 2022). Ideally, a single feature would correspond to a single neuron. However, in real-world neural networks, a major challenge is **polysemanticity** – the phenomenon where a single neuron encodes multiple features to accommodate the model's limited number of neurons (Olah et al., 2020; Elhage et al., 2022). A polysemantic neuron responds to different, often unrelated concepts, undermining the mechanistic approach to AI. Polysemanticity is explored in detail in Sect. 4.3.

MI seeks to interpret the connections and interactions between features by analyzing **circuits** – features linked by weights. Circuits are the minimal computational subgraphs of neural networks that are responsible for performing a given task. They are defined as directed, acyclic graphs. For instance, the curve detector feature is linked with downstream

neurons to represent more complex features (Olah et al., 2020). Formally, in a neural network represented as a directed graph $G = (V, E)$, where V is the set of nodes (neurons in the network) and E is the set of weighted edges (representing weighted connections between neurons), a circuit is a subgraph within G : $C = (V_C, E_C)$, where $V_C \subseteq V$ and $E_C \subseteq E$, that encapsulates a coherent computational process at various levels of abstraction.

Circuit discovery is the leading task defining mechanistic interpretability, and has already demonstrated promising results in both smaller and more complex models (Lieberum et al., 2023). The study of circuits could pave the way for proving the universality hypothesis – the idea that similar circuits and features are present across tasks and models. Such recurring patterns are **motifs**. The universality hypothesis, if true, may allow for knowledge transfer from small, toy models to state-of-the-art large-scale models (Chughtai et al., 2023; Merullo et al., 2024; Nainani et al., 2024).

Taxonomy of MI approaches. Mechanistic interpretability approaches can be organized based on different levels of analysis. Although all methods are united by the "top-down approach" aimed at unfolding input–output mechanisms responsible for a model's behaviors, they employ different techniques, and the definition of the entities analyzed varies across studies. The proposed taxonomy divides MI approaches according to three criteria: scope, task, and nature of analysis (Fig. 2).

Scope. Scope refers to the part of neural network being interpreted. Any component involved in the forward pass can serve as a unit of analysis (Rai et al., 2025). The narrowest level of analysis focuses on individual neurons, identifying their role in specific tasks – for instance, identifying a neuron crucial for a given decision (Olah et al., 2020). Researchers have increasingly generalized this analysis to more complex elements, such as attention heads or whole layers (Wang et al., 2022). Broader scope, such as circuit, refers to studying structures within a network, where the connections between components play a crucial role

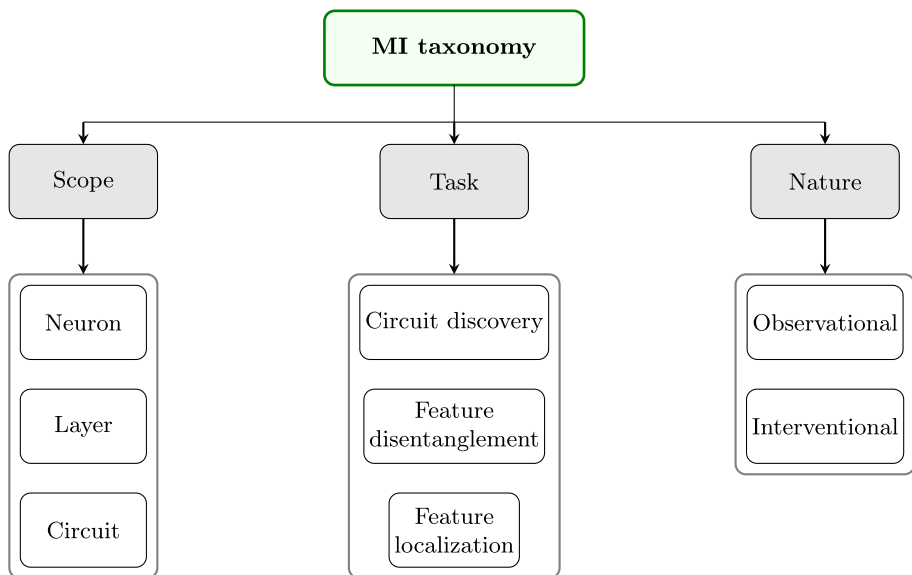


Fig. 2 Taxonomy of MI methods. The component and task criteria are not limited to the presented examples

in interpretation. At the broadest level, circuits – structures spanning multiple components – are studied as a whole. At this level, the connections between components, rather than the components themselves, play the crucial role in interpretation.

Task. The primary objective of MI is to reverse-engineer a neural network; however, most of the current research addresses only fragments of this goal. Among these are tasks such as feature localization, circuit discovery, and feature disentanglement. Feature localization refers to identifying components that are responsible for specific features. Circuit discovery is the task of uncovering meaningful structures in the network. Feature disentanglement focuses on resolving polysemanticity. While all contribute valuable pieces to the puzzle, they fall short of providing a holistic understanding of neural networks. Researchers often combine multiple methods to obtain a more complete understanding (e.g., validating a circuit found with patching via probing or visualization (Marks & Tegmark, 2024)).

Nature of analysis. MI can be divided into two main categories: observation-based approaches and intervention-based approaches. The key difference lies in whether the focus is on passively observing the model's behavior or actively modifying its components to study the effects. Observation-based approaches concentrate on examining the internal mechanisms of a neural network without making any changes to its architecture or parameters. These methods typically involve techniques such as visualizing weights, activations, or gradients, as well as tracing the flow of information through the network. Conversely, intervention-based approaches involve actively altering the network or its inputs to investigate causal relationships and gain insights into specific behaviors.

Comparison of MI approaches.

Despite the rapid growth of mechanistic interpretability, direct comparisons between techniques remain rare (Mueller et al., 2025a). Most studies evaluate a single method or a small set of closely related variants, on an ad-hoc task of their own choosing, making it difficult to assess whether progress reflects genuine methodological improvement or differences in evaluation setup. In principle, methods as different as probing, activation patching, and sparse autoencoders could be evaluated against the same model and task using shared metrics, but such cross-method comparisons have only recently begun to emerge (Mueller et al., 2025a).

Recent benchmarking efforts have started to address this gap. Mueller et al. (2025b) introduce the Mechanistic Interpretability Benchmark (MIB), which evaluates circuit discovery and causal variable localization methods across multiple tasks and models. The benchmark includes well-studied tasks such as indirect object identification (Wang et al., 2022) and arithmetic (Stolfo et al., 2023). Gupta and Kwa propose InterpBench (Gupta et al., 2024), which instead evaluates methods on synthetic transformer models with known ground-truth circuits, avoiding the lack of verified circuits in naturally trained models. However, to the best of our knowledge, the field still lacks interpretability benchmarks for other model families, such as CNNs.

In this survey, we do not conduct an experimental evaluation; instead, we synthesize existing findings into a structural comparison of methods' high-level properties. Table 2 compares techniques along the proposed criteria (task, scope, nature) and summarizes their main advantages and disadvantages, providing the kind of cross-method overview that, as discussed above, remains largely absent from the empirical literature. Performing an experimental comparison at the current stage of the field would likely compare experimental protocols rather than the underlying methods themselves. The absence of standardized

empirical comparisons should not be interpreted as a weakness of this survey, but rather as a reflection of the current state of mechanistic interpretability.

3 The Rise of Mechanistic Interpretability

We conducted a comprehensive literature review to illustrate the rapid emergence and growth of mechanistic interpretability, highlighting key trends and the increasing volume of research in recent years. All search phrases were enclosed in quotation marks.

Methodology. At the start of our research, we performed an initial search in the *Web of Science* database without time restrictions, aiming to situate MI within the broader context

Table 2 Mechanistic interpretability approaches, their causal nature, and trade-offs

Technique	Scope	Nature	Pros	Cons	Papers
Probing	Neuron, Layer	Observational	Simple to implement; works on frozen models; scales to large datasets	Correlational only – presence of decodable information does not imply the model uses it; requires labeled concepts	(Alain & Bengio, 2017; Gurnee et al., 2023; Gurnee & Tegmark, 2024; Chowdhury et al., 2025)
Lenses	Layer	Observational	Cheap, no extra training; reveals where predictions emerge across depth	Assumes residual stream is linearly decodable into vocabulary space; can be misleading for early layers	(2020), (Belrose et al., 2023; Katz & Belinkov, 2023; Dreyer et al., 2025)
Activation Patching	Neuron, Layer, Circuit	Interventional	Establishes causal effect of an activation on output; flexible, fine-grained	Computationally expensive at scale; choice of corrupted/clean baseline affects results	(Stolfo et al., 2023; Vig et al., 2020; Conmy et al., 2023; Heimersheim & Nanda, 2024)
Path Patching	Circuit	Interventional	Isolates causal contribution of specific computational paths, not just nodes	More complex to implement; combinatorial cost grows with circuit size	(Wang et al., 2022; Goldowsky-Dill et al., 2023; Meng et al., 2022)
Ablation	Neuron, Layer, Circuit	Interventional	Simple causal test; minimal assumptions; widely applicable	Can push activations out-of-distribution, producing misleading effects; coarse compared to patching	(Nanda et al., 2023; Bhaskar et al., 2024; Li et al., 2024; Garcia-Carrasco et al., 2024)
Causal analysis	Circuit	Interventional	Rigorous hypothesis-testing framework; controls for spurious correlations in circuit claims	High implementation complexity; computationally costly; requires precise hypothesis specification	(Chan et al., 2022; Jenner et al., 2023; Geiger et al., 2024)
Sparse Autoencoders (SAEs)	Neuron, Layer	Observational	Discovers monosemantic features without labeled data; scales to large models	No ground-truth validation of recovered features; reconstruction-vs-interpretability trade-off; feature splitting/absorption issues	(Huben et al., 2024; He et al., 2024; Marks et al., 2024; O’neill & Bui, 2024)

of research on AI explanation and interpretability. Our initial search indicated that publications specifically focused on mechanistic interpretability only began to appear around 2015, meaning earlier related work would not have been captured under MI-specific terminology. We therefore examined the wider field using the following keywords:

- *Explainable Artificial Intelligence* (similar results with *Explainable Artificial Intelligence OR Explainable Machine Learning*): One publication appeared in 2004, 2005 and 2006; two in 2017, and 22 in 2018, followed by rapid growth.
- *Interpretable Artificial Intelligence OR Interpretable Machine Learning*: The first publication appeared in 2006; publication growth remained slow until 2018 (33 works), after which it accelerated. For *Interpretable Machine Learning* alone, the number rose from 4 works in 2014 to 960 in 2024.
- *Mechanistic Interpretability AND Machine Learning*: isolated publications appeared between 2010 and 2022; In 2024, 23 publications were listed.

Based on this and additional database analyses, we decided to focus our quantitative analysis on the period since 2015, reflecting the dynamic expansion of AI model interpretation methods.

The literature search was conducted in three databases: *Scopus*,¹ *Web of Science*,² and *PubMed*.³ We encountered two major challenges when researching the mechanistic interpretability of neural networks:

- *Exclusion*: referring to papers related to mechanistic interpretability, but not mentioning their mechanistic background directly.
- *Breadth*: referring to papers related to mechanistic explanations in general but not to AI models.

Exclusion results from the novelty of MI in the context of AI interpretability. Many researchers who study topics related to mechanistic interpretability may not be aware of this naming convention. Moreover, as exemplified with probing, some approaches are now considered mechanistic despite their different origins. In addition, much of the early MI research has been published in blogs and forums, thus it is not listed in research databases.

Breadth stems from the philosophical and scientific background of MI. Mechanistic explanations are widely used across various disciplines, including biology, physics, and engineering. This broad applicability makes it difficult to filter and curate AI-specific literature, as many papers discuss mechanistic insights without being directly relevant to neural network interpretability. Consequently, the search term "mechanistic interpretability" often yields results not related to neural network interpretability.

We studied several search terms, including "mechanistic interpretability" which suffered from breadth, "mechanistic interpretability AND neural networks", which suffered from exclusion. Ultimately, we limited the scope of the search to "Mechanistic interpretability AND Machine Learning". This query both captured the literature related to interpretability

¹ <https://www.scopus.com>.

² www.webofscience.com/wos/.

³ <https://pubmed.ncbi.nlm.nih.gov>.

of deep neural networks and excluded works related to mechanistic explanations in other disciplines.

The problem of broadness was especially prevalent in PubMed. The majority of papers were related to mechanistic explanations in medicine, for instance (Gaw et al., 2019), where the authors combine machine learning with mechanistic models to study glioblastoma. Distinguishing AI-focused mechanistic interpretability papers from the broader landscape of mechanistic explanations would require careful manual selection. Conversely, results from Web of Science manifested exclusion and yielded few publications when searching for "Mechanistic interpretability" (24 papers on April 17th). The results were accurate, yet sparse because of this exclusion.

Results of literature search. We present the results of our literature search conducted in Scopus. For clarity, we limited the presentation to one database due to the issues described previously. We limited the search to the period 2015–2025 and included preprints. First, we searched for papers on explainable, interpretable and trustworthy AI. A clear upward trend can be observed that can be associated with the rise of transformer-based models (Lin et al., 2022). Detailed results are presented in Fig. 3.

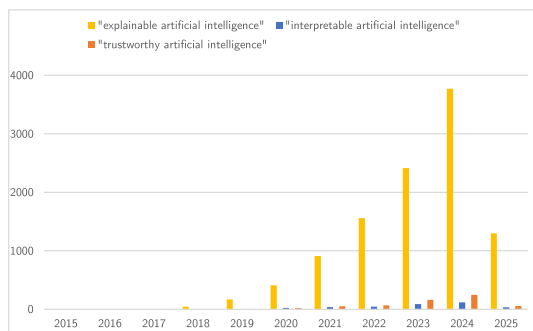
As depicted in Fig. 3, researchers have made significant progress by expanding the XAI toolbox (Longo et al., 2024) in recent years. However, interpretable and trustworthy AI constitute only a fraction of the publications.

We observe a significant dominance of publications on *Explainable AI* over the others. The finding is expected, as this term has the broadest meaning and is the most popular in the research community. The narrowest term is *Trustworthy*, a concept that has gained popularity recently, and research in this area is still in its early stages. *Interpretable AI* often refers to self-explaining models, while there is a high demand for the interpretability of black-box models.

A similar situation is observed in the *Web of Science* database. Publications related to "explainable artificial intelligence" clearly dominate: as of May 8, 2025, there were 1,963 publications recorded for 2024 (compared to 1,274 in 2023 and 934 in 2022). The keyword "trustworthy artificial intelligence" returned 76 publications in 2024, compared to 58 in 2023 and 13 in 2020 (with no records in earlier years). The keyword "interpretable artificial intelligence" returned 64 publications in 2024, compared to 51 in 2023, 12 in 2020, and one in 2019. The *PubMed* database was not included in the analysis, as it would require manual verification of the retrieved results.

Figure 4 presents the number of publications referring to MI each year since 2015. The notable growth in publications since 2020 coincides with the introduction of the term

Fig. 3 Number of total publications and preprints referring to explainable, interpretable, and trustworthy AI since 2015. Data retrieved from Scopus (April 17th, 2025) by using the search terms: explainable artificial intelligence, interpretable artificial intelligence, trustworthy artificial intelligence



"mechanistic interpretability" in the context of neural networks. The difference in publication numbers for the two terms is a result of broadness. In the *Web of Science*, the keyword "mechanistic interpretability" returned 23 publications in 2024 (compared to 9 in 2023 and 3 in 2022). The keywords "mechanistic interpretation" AND "machine learning" returned only five publications in 2024, two in 2023, and one publication each in 2022, 2021, 2020, and 2018.

Interest in mechanistic interpretability is currently limited compared to other areas of XAI research, as it requires significant expertise and effort to dissect complex models at a granular level. However, as depicted in Fig. 4, this field is rapidly gaining traction due to its potential to provide deeper insights into the inner workings of machine learning systems. Recent work has shown that MI techniques can scale beyond toy settings – for instance, sparse autoencoders have been successfully applied to extract interpretable features from production-scale language models (Templeton et al., 2024), and circuit-level analyses have identified concrete computational subgraphs underlying specific behaviors in widely studied models (Wang et al., 2022). Such results suggest that MI is a promising avenue for addressing challenges in AI alignment, debugging, and ethical deployment. While still an emerging discipline, its ability to deliver detailed insights positions it as a critical component of the future of XAI research.

4 Current State of MI Research

Mechanistic interpretability techniques encompass a diverse and evolving range of approaches aimed at understanding the inner workings of neural networks. These techniques range from narrow analyses, such as visualizing individual neurons and their activations, to

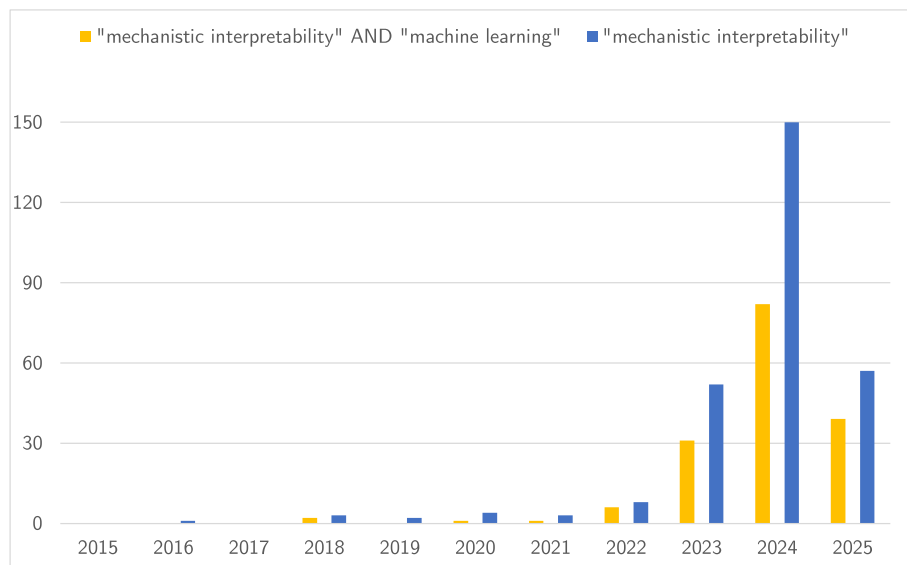


Fig. 4 Number of total publications and preprints referring to mechanistic interpretability since 2015. Data retrieved from Scopus (April 17th, 2025) by using the search term: "mechanistic interpretability" and "mechanistic interpretability" AND "machine learning"

approaches that examine the relationships between entire layers or components, such as circuits or attention heads. Based on the available literature, we identify key milestones in the development of MI, summarized in Table 3. The first column references the literature; the remaining columns summarize the task addressed, the main contribution, the model studied (where applicable), and the year of publication.

Building on the taxonomy introduced in the previous section, we present the main techniques associated with each interpretability task. Each technique is presented with a conceptual description of its foundations, a mathematical formalization where applicable, and concrete examples illustrating its application in real-world neural networks.

Given the scale and rapid pace of growth of mechanistic interpretability research, our goal was not to provide an exhaustive catalogue of every published method. Instead, we selected representative techniques that are widely recognized as foundational within the field or that introduced a distinct methodological direction subsequently adopted by later work. The scope of this survey focuses primarily on methods developed for transformer-based architectures, as these currently constitute the dominant body of mechanistic interpretability research, while emerging work on other model families is discussed where relevant.

Table 3 Milestones in mechanistic interpretability

References	Task	Main contribution (model studied)	Year
Olah et al. (2020)	Circuit discovery	Introduction of circuits concept (InceptionV1)	2020
Vig et al. (2020)	Circuit discovery	Application of CMA to circuits (GPT-2)	2020
Elhage et al. (2021)	Reverse-engineering	Mathematical framework for transformers (toy transformer)	2021
Geiger et al. (2021)	Hypothesis verification	Causal abstraction analysis (BiLSTM, BERT-based model)	2020
Wang et al. (2022)	Circuit discovery	Introduction of path patching method (GPT-2 small)	2022
Elhage et al. (2022)	Superposition study	Demonstration of superposition (toy models)	2022
Sharkey and Braun (2022)	Feature disentanglement	Recovery of ground truth features (small transformer)	2022
Conmy et al. (2023)	Circuit discovery	Introduction of ACDC method (GPT-2 small)	2023
Gurnee et al. (2023)	Locating features	Sparse probing technique (Pythia models)	2023
He et al. (2024)	Circuit discovery	Dictionary learning for circuits (Othello-GPT)	2023
Geiger et al. (2024)	Other	Causal abstraction framework for MI	2023

4.1 Feature Localization

Feature localization refers to identifying and characterizing individual neurons that encode some *important* information. In ML, the common characteristic of a neuron is its *importance* – a measure of their contribution to the model’s prediction. Importance is usually quantified as the activation or gradient value for a given input (Dhamdhere et al., 2018). Studying individual neurons is the first step to reverse-engineer a neural network, as they can be thought of as variables in the encoded algorithm.

4.1.1 Probing

Probing is a technique derived from NLP to interpret and analyze the internal representations learned by models (Alain & Bengio, 2017). This method is based on the premise that neural networks do not acquire any new information during their forward passes but rather transform given inputs to learn increasingly meaningful representations. By training a simple classifier, called a *probe*, on these generated representations, researchers can determine whether the network encodes any specific type of information (Belinkov, 2022). Figure 5 provides a schematic overview of probing.

Mathematically, probing can be framed as a supervised learning problem (Hewitt & Liang, 2019). Given a model f that generates hidden representations $\mathbf{h}_i \in \mathbb{R}^d$ for each input token or sequence x_i , the goal is to train a lightweight probe g that maps $\mathbf{h}_i$ to a target property y_i (e.g., part-of-speech tag, dependency label). Formally, the probe learns a function $g : \mathbb{R}^d \rightarrow \mathcal{Y}$, where $\mathcal{Y}$ is the label space, by minimizing a loss function, $\mathcal{L}(g(\mathbf{h}_i), y_i)$. The performance of the probe g on a held-out test set measures how well the representations $\mathbf{h}_i$ encode the property y_i (Alain & Bengio, 2017; Belinkov, 2022). Algorithm 1 presents the pseudo-code of linear probing.

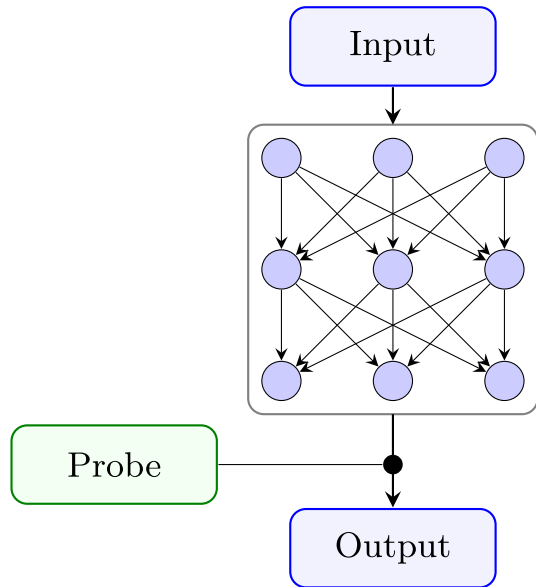
Require: Pretrained model f , dataset $D = \{(x_i, y_i)\}_{i=1}^n$, target layer L , loss function $\mathcal{L}$

- 1: Initialize empty set $\mathcal{H} \leftarrow \emptyset$
- 2: **for** each $(x_i, y_i) \in D$ **do**
- 3: $\mathbf{h}_i \leftarrow f_L(x_i)$ ▷ Extract representation from layer L
- 4: Add $(\mathbf{h}_i, y_i)$ to $\mathcal{H}$
- 5: **end for**
- 6: Define probe $g : \mathbb{R}^d \rightarrow \mathcal{Y}$ ▷ e.g., linear classifier
- 7: Train g on $\mathcal{H}$ by minimizing $\mathcal{L}(g(\mathbf{h}_i), y_i)$
- 8: Evaluate g on a held-out test set
- 9: **return** Performance metrics (e.g., accuracy, F1)

Algorithm 1 Linear probing of internal representations

In language models, probing is used to assess whether the model has captured any linguistic knowledge. For instance, Conneau et al. (2018) introduce 10 probing tasks for single sentence embeddings and Lin et al. (2019) probe BERT’s (Devlin et al., 2019) representations to identify structurally-defined elements. Gurnee and Tegmark (2024) employ probing to identify neurons containing spatial and temporal features, and also provide evidence that validates the linear representation hypothesis (Sect. 4.3).

Fig. 5 Illustration of probing. The probe does not interact with the neural network; rather it uses its representation to yield explanations. Note that probes can be applied at any layer of the network



In a mechanistic approach, probing is aimed at identifying whether and where the model encodes particular kinds of information, thereby complementing the mechanistic understanding (Bereska & Gavves, 2024). Probing can identify the roles of individual neurons and the features represented in the network, aiding in the localization of information (Gurnee et al., 2023). In particular, **sparse probing**, which attempts to identify only a subset of neurons (or a single neuron) that is activated by a given input. This is achieved by limiting the classifier to k non-zero coefficients, thus creating a k -sparse probe. Sparse probing can be viewed as a neuron ranking problem – scoring individual neurons based on their importance in a given task. It filters out irrelevant neurons, highlighting the specific representations that encode linguistic or functional knowledge. However, due to the focus on sparsity, important neurons can be overlooked (Gurnee et al., 2023; Antverg & Belinkov, 2022).

Gurnee et al. (2023) propose two techniques: adaptive thresholding, which constrains the value of k through iterative retraining, and optimal sparse probing which employs a cutting plane algorithm for smaller values of k . Chowdhury et al. (2025) employ ridge regression-based probes to identify context neurons involved in the task of document ranking. The authors use Lasso regularization to enforce sparsity in the classifiers. These probes are applied across the layers of the RankLlama model (Ma et al., 2024), revealing multiple statistical features captured along the layers of the network.

Other examples of probing include **edge probing** and **structural probing**, which are complementary methods used to evaluate linguistic knowledge encoded in language model representations, differing in focus and scope. Edge probing assesses whether local pairwise relationships between words or tokens (edges) in a linguistic structure are encoded within the representations of a model. Rather than classifying individual token properties, edge probes focus on the relationships or interactions between spans, such as syntactic dependencies or semantic roles (Tenney et al., 2019; Choudhury & Kalra, 2023).

In contrast, structural probing evaluates the encoding of global hierarchical structures, such as full syntactic dependency trees or semantic frames. Structural probes examine how

well linguistic structures are embedded in model representations, capturing the relationships between all elements in a sentence. For example, Hewitt and Liang (2019) present a structural probe to identify syntax trees embedded within the representations of ELMo (Peters et al., 2018) and BERT (Devlin et al., 2019) by predicting distances or reconstructing hierarchical tree structures. Structural probing provides insight into the broader organization of linguistic information encoded in embeddings. However, the self-supervised nature of structural probes raises doubts as to about whether the probe actually captures any knowledge of language models (Farquhar et al., 2023).

Taken together, probing methods offer a practical and scalable entry point into mechanistic analysis, requiring minimal intervention on the model. Nevertheless, their observational nature means that high probe accuracy reflects correlation rather than causation – a fundamental limitation that motivates the interventional approaches presented in the following sections.

4.1.2 Lenses

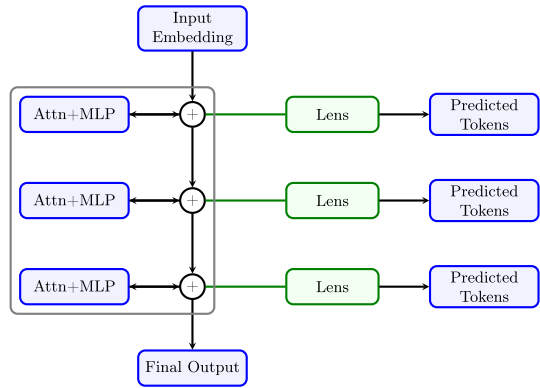
In contrast to probing, lenses do not focus solely on observing a specific state of the model. Rather, they map the representations from intermediate layers to the model's vocabulary distribution, without passing them through the remaining layers of the model. By revealing how the model's predictions develop across the network, this approach makes it possible to investigate what linguistic or semantic information is encoded at different stages of the model's computation (Elhage et al., 2021; Geva et al., 2021). Note that lenses are limited to transformer architectures, as they rely on projecting residual stream activations into vocabulary space. Figure 6 depicts multiple lenses applied at different layers of a transformer model.

Formally, a lens maps the hidden state h^l at layer l to the vocabulary space by multiplying it with the output embedding matrix, W_{out} , yielding the vocabulary logits, z_l (Rai et al., 2025):

$$z_l = W_{out}h^l$$

The **logit lens**, first introduced by nostalgebraist (2020) in 2020, was quickly adopted by researchers and applied in various transformer architectures (Luo & Specia, 2024). Algorithm 2 presents the pseudo-code of the lens. Early works focused on analyzing only specific elements of transformer architecture: Geva et al. (2021, 2022) analyze how feed-forward layers update token representations, enabling these representations to be interpreted at any stage of the model's computation, and Sakarvadia et al. (2024) map attention heads to the vocabulary space and identify a mechanism responsible for retrieving memories. Based on the attention module's memory, the lens has also been employed to model the information flow in GPT-2 (Katz & Belinkov, 2023). Dar et al. (2023) build upon the logit lens by taking into account all weights of a model and propose a framework for interpreting model's parameters in isolation. Merullo et al. (2024) employ the logit lens in an in-context learning setting, uncovering a mechanism that implements simple vector arithmetic. A different application of the lens was demonstrated by Wendler et al. (2024), who investigated whether the representations of the Llama-2 model Touvron et al. (2023) consistently use English as their pivot language regardless of the prompt.

Fig. 6 Illustration of lenses. A lens can be applied at any stage of the residual stream and project the intermediate embeddings to tokens



Require: Model f , output embedding matrix $\mathbf{W}_{out}$, input sequence x , layer indices $\mathcal{L} = \{1, 2, \dots, L\}$

- 1: Tokenize input: $x \rightarrow (x_1, x_2, \dots, x_T)$
- 2: Pass input through model to extract hidden states $\{h^l\}_{l \in \mathcal{L}}$
- 3: **for** each layer $l \in \mathcal{L}$ **do**
- 4: Compute vocabulary logits: $\mathbf{z}_l = \mathbf{W}_{out} h^l$
- 5: Store or visualize top-k predictions from $\mathbf{z}_l$
- 6: **end for**
- 7: **return** Layer-wise vocabulary predictions $\{\mathbf{z}_l\}_{l \in \mathcal{L}}$

Algorithm 2 Logit lens

An extension of the logit lens is the **DecoderLens**, which overcomes its limitations in interpreting representations within encoder-decoder models. Langedijk et al. (2024) propose a framework in which the decoder can cross-attend to intermediate encoder layer representations, rather than relying solely on the final encoder output, providing richer insights into the model's internal workings.

Other techniques include the **linear lens**, which questions the idea of mapping hidden representations directly to the output distribution. Din et al. (2024) project hidden representations between layers by fitting a linear regression from source layer l_i to target layer l_{i+1} . Similarly, the **tuned lens** (Belrose et al., 2023) learns affine transformations that *translate* representations from an internal layer to the final layer. The **future lens** investigates whether future tokens can be predicted from hidden states. To achieve this, Pal et al. (2023) train a linear model f_θ transforming a hidden encoding at layer l , h_T^l , to a future hidden state at layer L , h_{T+1}^L . A further example is the **JailbreakLens**, an interpretation framework introduced by He et al. (2025), which investigates how jailbreak attacks impact LLM's representations. Lenses have also been applied beyond LLMs; for instance, the **diffusion lens** employed in text-to-image diffusion models (Toker et al., 2024) and the **SemanticLens** – a universal explanation tool which maps neural network components into a semantically structured space (Dreyer et al., 2025).

Overall, lens-based methods offer a complementary perspective on model representations by tracing how information evolves across layers. They extend the observational logic of probing by removing the need for labeled concepts. Their main advantage lies in their ability to inspect intermediate representations without an additional correlational model. Their limitation – that they assume linear decodability at every layer – also highlights a

broader challenge in MI: the tools we use to observe a model's computations inevitably carry their own assumptions about how those computations are structured.

4.2 Circuit Discovery

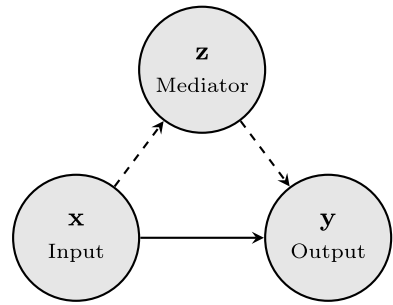
Circuit discovery aims to identify causal graphs in neural networks – structured subnetworks of neurons and connections that encode specific computational functions. This process supports the view that neural networks implement algorithmic computations, thereby reinforcing the idea of mechanistic interpretability and moving beyond black-box explanations. Empirical studies have demonstrated the emergence of structured circuits across neural networks, both in small toy models (Nanda et al., 2023; Zhong et al., 2023; Kitouni et al., 2024) and large-scale real-world networks (Lieberum et al., 2023; Wang et al., 2022). These findings reinforce the view that neural networks, despite their complexity, operate through interpretable and decomposable computational structures rather than relying solely on distributed, uninterpretable representations.

The principal framework for circuit discovery is based on Causal Mediation Analysis (CMA). CMA is a method derived from causal inference used to study systems of dependencies, typically represented by directed acyclic graphs (Pearl, 1995). It builds upon the causal framework by including *mediators* in its analyses – intermediate elements in a system that influence the final outcome. In neural networks, causal mediation analysis can be used to investigate circuits by treating internal model components (e.g., neurons) as mediators between inputs and outputs (Vig et al., 2020). Figure 7 illustrates a causal graph with a mediator.

The core method derived from CMA in mechanistic interpretability is **patching** (also known as interchange intervention Geiger et al. 2024) – a method employed to study and understand the internal computations of a machine learning model by modifying – patching – its internal components during inference. The objective is to identify specific components of the model (e.g., particular neurons, attention heads, layers) that are causally responsible for certain behaviors or outputs. Patching stands as a fundamental approach in MI, as it is by definition not limited to studying only the input–output relations but emphasizes the importance of internal components (Vig et al., 2020). Patching aims to uncover computational circuits by intervening in model components and observing the resulting changes in output. This technique views models as causal graphs in which internal components (such as neurons) form causal pathways connecting inputs to outputs (Stolfo et al., 2023; Vig et al., 2020; Geiger et al., 2021; Finlayson et al., 2021).

4.2.1 Activation Patching

Activation patching is the primary technique derived from causal interference. The activation-based approach focuses on modifying or replacing the activations of neurons within a neural network model. Its primary use is to investigate the role of specific neurons or layers in the model's decision-making process by observing how the changes in activations impact the model's output (Bereska & Gavves, 2024). Activation patching follows a standard workflow to identify crucial activations in neural networks (Heimersheim & Nanda, 2024):

Fig. 7 Causal mediation analysis

1. Clean run – the model processes a clean input, generating a correct prediction, and caches the activations.
2. Corrupted run – the model processes a corrupted input, so that the prediction is impaired.
3. Patched run – the model processes a corrupted input, but selected activations are substituted with those from the clean run.

The workflow is presented in Fig. 8, which depicts two forward passes: one processing a clean input, and the other processing a corrupted one. Activations are patched from the clean pass into the corrupted in order to restore the output to its clean state. Algorithm 3 presents the pseudo-code of activation patching.

Require: Model f , clean input x_{clean} , corrupted input x_{corrupt} , layers $\mathcal{L}$, positions $\mathcal{T}$, selected components $\mathcal{C}_t^l$

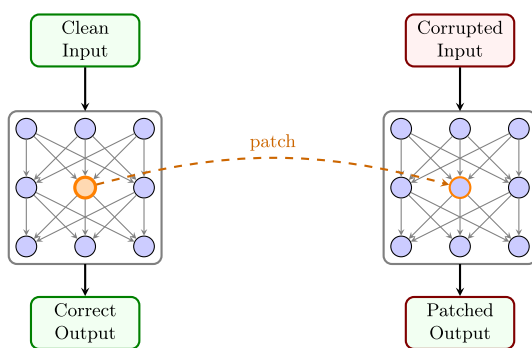
- 1: Run forward pass on x_{clean} to obtain activations $h_{\text{clean}}^l[t]$
- 2: Run forward pass on x_{corrupt} to obtain activations $h_{\text{corrupt}}^l[t]$
- 3: **for** each component $c \in \mathcal{C}_t^l$ **do**
- 4: In the corrupted run, replace $h_{\text{corrupt}}^l[t] \leftarrow h_{\text{clean}}^l[t]$
- 5: Continue the forward pass from layer $l + 1$ onward using patched activations
- 6: Record model output after patching (e.g., logits or probabilities)
- 7: Compare patched output with clean and corrupted outputs
- 8: **end for**
- 9: **return** Degree of restoration (e.g., logit difference, KL divergence)

Algorithm 3 Activation patching

Based on the model's performance during the patched run, researchers can determine the significance of the replaced components, narrowing down a computational circuit. Conmy et al. (2023) propose a method for automating the workflow – ACDC (Automated Circuit Discovery), an iterative algorithm that evaluates and patches activations at each node so that the performance of the model remains unaffected. However, in both manual and automated approaches, the substitutions of activations are performed iteratively, resulting in a substantial computational cost of this technique. Other automation techniques include more accurate and scalable ACDC, named ACDC++ (Syed et al., 2023) and Contextual Decomposition for Transformers (CD-T) (Hsu et al., 2024).

Activation patching has been used to identify critical activations for storing and processing data; for instance, Stolfo et al. (2023) find that language models process arithmetic information by transmitting relevant data from early to late layers using the attention mechanism. This information is then processed by a set of MLP modules, which generate the final information relevant to the final prediction incorporated into the residual stream. Meng et al. (2022) employ activation patching to investigate the storage and recall of factual associa-

Fig. 8 Illustration of activation patching. The patching direction depicted is clean $\rightarrow$ corrupted, meaning the procedure attempts to find the minimal number of neurons that will restore the output



tions within autoregressive transformer language models. Other researchers investigate circuits in tasks such as syntactic agreement (Finlayson et al., 2021) or multiple-choice question answering (Lieberum et al., 2023). Furthermore, by employing activation patching, Todd et al. (2024) find a key mechanism of in-context learning: function vectors, which trigger the execution of specific procedures in a language model. Marks and Tegmark (2024) employ activation patching to locate the representations of truth in the LLaMA-2 model, and further analyze these hidden states through visualization and probing techniques. Lan et al. (2024) investigate motifs through activation patching and demonstrate a circuit responsible for sequence continuation tasks. Monea et al. (2024) enhance activation patching by focusing on corrupting tokens rather than embeddings. Beyond language models, activation patching was applied by Palit et al. (2023) to study visual question answering in BLIP (Li et al., 2022). Their results suggest that the vision modality is located in the final layers of the model.

4.2.2 Attribution Patching

Attribution patching emerged as a response to the computational complexity of the standard procedure of activation patching (Nanda, 2023). Instead of performing a forward pass for each substitution, this approach uses a gradient-based approximation to find a linear estimate between the corrupted and clean pass. Attribution patching allows researchers to identify the circuit in only two forward passes and one backward pass, while simultaneously reducing the need for human supervision during the process (Ferrando & Voita, 2024). Syed et al. (2023) extend this approach by proposing **edge attribution patching**, an automated circuit discovery algorithm that obtains the most important edges in a given task. Hanna et al. (2024) further build upon edge attribution patching by integrating gradients to increase the faithfulness of the discovered circuits.

4.2.3 Path Patching

Path patching constrains the interventions to a specific path in the model, leaving the rest of the network untouched. The goal is to determine whether the patched path causally contributes to a specific behavior or output. In contrast to other patching methods, path patching investigates the interactions of components in a network on each other (Hanna et al., 2023). Wang et al. (2022) show its application in discovering a circuit in GPT-2

small. Goldowsky-Dill et al. (2023) generalize the idea for any number of paths, modeling the network as an arbitrary computational graph.

Activation patching and its variants (attribution, path) represent a significant step toward causal rather than merely correlational interpretability. This causal framing is what distinguishes the patching family from probing and lens-based approaches. Nevertheless, though the idea of localizing important components in neural networks through interventions seems sensible, researchers have questioned the faithfulness of such interpretations. Even when an intervention measurably affects model response, it is not sufficient to clearly establish that the targeted component plays the specific causal role attributed to it Wang and Veitch (2024). The effect may instead reflect confounding factors, distributed redundancy, or the intervention pushing the model out of its training distribution. Hase et al. (2023) question the reliability of mechanistic understanding in locating model behavior. Their experiments show that successfully editing a behavior at a given location does not confirm that this location is where the original computation actually occurs. Current research aims to increase both the interpretability and the faithfulness of circuits by optimizing intervention techniques (Wang & Veitch, 2024; Stoebr et al., 2024). Reliable validation of discovered circuits remains one of the central open challenges of the field.

4.2.4 Ablation Study

Ablation, also known as knockout, is another technique derived from causal inference. Although both ablation studies and patching involve the manipulation of a system's components, they differ in their specific approaches and objectives. Ablation is primarily focused on removing or disabling certain elements within the system to observe the resulting changes in behavior, thus identifying which components are essential for the functionality of the system (Meyes et al., 2019). In contrast to patching, ablation does not seek to understand the causal relationships between components; but rather their causal importance (Lan et al., 2024; Li et al., 2024). The removal of components is usually performed via *zero ablation*, which sets their values to zero, and *mean ablation*, which replaces their values with the average values across a training distribution. However, zero ablation often results in noisy results, as setting components to zero may introduce artificial disruptions that do not naturally occur in the model (Wang et al., 2022). A novel approach is *optimal ablation*, proposed by Li et al. (2024), that sets the value to a constant that yields the minimal loss of the ablated model.

Ablation has been applied to various components of neural networks. Ghorbani et al. (2020) introduced Neuron Shapley, a framework employing zero ablation to identify the crucial neurons. In their work, they quantify the importance score of neurons based on their estimated Shapley value, taking interaction between components into account (e.g., if two neurons are both required to improve the performance of the model). Alternatively, Olsson et al. (2022) study the causal importance of induction heads in in-context learning. Geva et al. (2023) and Fierro et al. (2025) knock out attention weights to investigate how GPT models extract factual knowledge. Li et al. (2024) focus on disabling edges responsible for bad behaviors in models (e.g., hallucinations) using **targeted edge ablation**, and García-Carrasco et al. (2024) combine ablation with the logit lens to detect vulnerabilities in GPT-2 Small Radford et al. (2019).

Another example of ablation study is **causal scrubbing**, a generalized approach that aims to test hypotheses about why a model behaves in a specific way. In contrast to other causal approaches, scrubbing begins with establishing a hypothesis about the circuits. The algorithm introduced by Chan et al. (2022) is as follows:

1. Define the components and pathways that the hypothesis claims are responsible for a mechanism.
2. Intervene on parts of the model that are not specified by the hypothesis.
3. Evaluate the output of the model – if the hypothesis is correct, it should remain unchanged.

In Chan et al. (2022), the interventions were applied using *resampling ablation*, which replaces the values of the components using any value they had on other inputs. Due to the restriction on the intervention space, causal scrubbing performs less interventions than other methods. However, it relies heavily on the hypotheses and can misidentify correlated features (Jenner et al., 2023). As such, causal scrubbing is best understood as a validation tool for hypotheses generated through less constrained methods, rather than a discovery tool in its own right.

Overall, circuit discovery represents the central methodological direction in mechanistic interpretability, shifting the focus from identifying isolated neurons toward understanding distributed computational pathways. While recent advances have substantially improved automation and scalability, reliable validation of discovered circuits remains one of the central open challenges of the field (Hanna et al., 2024; Hase et al., 2023).

4.3 Feature Disentanglement

Feature disentanglement methods aim to resolve the issue of polysemanticity, which represents a significant challenge to mechanistic interpretability. Ideally, a neuron would respond to a single feature, making the network monosemantic and easier to interpret. However, in practice, neurons are often observed to be polysemantic, meaning that they respond to multiple, often unrelated concepts (Goh et al., 2021; Olah et al., 2017). The primary reason for polysemanticity is the substantial number of features that the network attempts to accommodate within its limited number of neurons (Olah et al., 2020). Furthermore, Lecomte et al. (2024) highlight the phenomenon of *incidental polysemanticity*, where a network, despite having enough neurons, still overlaps representations of distinct features.

The *superposition hypothesis* provides a theoretical framework to explain this behavior, proposing that polysemanticity results from the *compression* of features into a limited representational space. The features are encoded not only in the n -dimensional activation space of a network but also in the almost orthogonal directions, as illustrated in Fig. 9. Elhage et al. (2022) demonstrate that superposition emerges naturally in neural networks. Further, Scherlis et al. (2025) show that polysemanticity depends on the allocated importance of features, i.e., important features are encoded orthogonally to other features. Other explanations behind polysemanticity include redundancy introduced during training (e.g., due to random dropout), or correlations in natural data that encourage neurons to share features (Chan, 2024).

The goal of feature disentanglement is to identify all the features encoded by the network, thereby providing a clearer understanding of its internal representations. Disentanglement

can be approached in two ways: ad-hoc, by designing models without superposition (Elhage et al., 2022), and post-hoc, by employing sparse coding to describe how features are represented in models with superposition (Sharkley et al., 2022).

4.3.1 Monosemanticity in Models

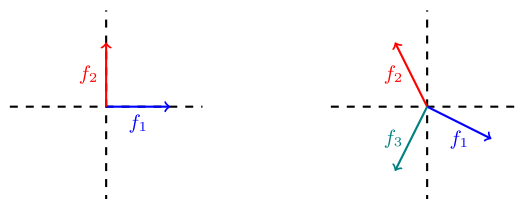
The motivation behind designing models with monosemantic neurons is their increased interpretability, as such neurons can be easily mapped to features. One possibility is the use of linear models, which do not exhibit superposition (Elhage et al., 2022). In non-linear networks, superposition can be reduced by decreasing the sparsity of features. However, this forces models to learn only the top n important features. Jermyn et al. (2022) engineer monosemanticity in toy models and show that increasing the number of neurons per layer reduces polysematicity, yet leads to increased computational cost. The benefits of monosemantic models are a subject of ongoing debate within the research community; Wang et al. (2024) demonstrate that decreasing monosemanticity benefits model performance, whereas Yan et al. (2024) show a positive correlation between monosemanticity and model capacity. The question whether monosemanticity should be inhibited or encouraged remains unsettled.

4.3.2 Sparse Autoencoders

Following the superposition hypothesis, activations can be decomposed as a linear combination of features (directions in the activation space) (Elhage et al., 2022; Park et al., 2024). Based on this premise, a natural solution to identifying individual features encoded by a network is the implementation of sparse coding (Olshausen & Field, 1997), especially sparse autoencoders (SAEs). Also known as sparse dictionary learning, the problem can be framed mathematically as follows: given an input dataset, $X = [\mathbf{x}_1, \dots, \mathbf{x}_i]$, $\mathbf{x}_i \in \mathbb{R}^m$, and a positive integer n , find a dictionary $\mathbf{D} \in \mathbb{R}^{m \times n}$, that allows for the sparsest representation $\mathbf{R} = [\mathbf{r}_1, \dots, \mathbf{r}_i]$, $\mathbf{r}_i \in \mathbb{R}^n$ so that $\|\mathbf{D}\mathbf{R} - \mathbf{x}\|_2$ is minimized (Tillmann, 2015).

In other words, SAEs transform an entangled space into a sparse representation, enabling independent analysis of features. Such decomposition offers a great opportunity for interpretability, as it allows the analysis of the network's components in isolation. SAEs have been successfully employed for the visualization of word embeddings in LLMs (Arora et al., 2018; Panigrahi et al., 2019; Subramanian et al., 2018; Yun et al., 2021), and more recently to extract features from LLMs (Huben et al., 2024; He et al., 2024; Deng et al., 2023). In the context of mechanistic interpretability, SAEs have been integrated as part of circuit discovery pipelines (He et al., 2024; O'Neill & Bui, 2024; Ge et al., 2024; Marks et al., 2025). Taken together, these findings indicate that sparse autoencoders currently represent the most promising direction for disentangling polysemantic representations.

Fig. 9 Superposition of features in two-dimensional space. Left: feature directions are orthogonal, no polysematicity. Right: Overlapping (almost orthogonal) feature directions, polysematicity



SAEs are not the only approach to feature disentanglement. O'mahony et al. (2023) proposed disentangling polysemantic neurons into concept vectors, which are obtained from intermediate representations of images that have the highest activation values for a neuron. Dunefsky et al. (2024) studied *transcoders*, a variant of SAEs which aims to reconstruct the original layer output, and demonstrated its use in circuit discovery.

Taken together, sparse autoencoders occupy a unique position in the MI landscape: unlike the methods discussed above, they operate without labeled concepts or pre-specified hypotheses, instead deriving candidate features directly from the model's own activations through unsupervised learning. This makes them particularly powerful for exploratory analysis at scale, though the absence of ground-truth features means that the interpretability of recovered directions must still be validated through human inspection or downstream causal testing. For instance, Chanin et al. (2025) show that as SAEs scale up, hierarchically related features tend to split into finer sub-features, but this splitting is unstable. Seemingly monosemantic latents can fail to fire where expected, with their activation instead absorbed by more specific child features – a failure mode unresolved by simply adjusting SAE size or sparsity. Such failures illustrate why recovered features cannot be taken at face value without further validation.

5 Challenges and Opportunities

Despite its potential, mechanistic interpretability still encounters numerous challenges. In this section, we address the major issues related to MI and outline some potential opportunities and applications.

5.1 Challenges

Mechanistic interpretability promises to fully uncover the computations of neural networks. However, many researchers have posed the question of whether achieving this ambitious goal is truly possible. Current research is largely limited to dissecting toy models or small fragments of larger neural networks. The narrow scope of MI research may lead to "cherry-picked" achievements, with methods lacking broader evaluation (Räuker et al., 2023). We outline the major challenges facing the field of mechanistic interpretability in Table 4, followed by a detailed description of the key problem areas.

Spurious explanations vs. genuine understanding. A key challenge in mechanistic interpretability is distinguishing between explanations that accurately reflect the model's internal computations and those that are merely artifacts of human-imposed narratives. Cherry-picked results alone are insufficient to understand neuron behavior. Currently, with no unified evaluation techniques, MI findings still rely heavily on human interpretation. This may lead to false assumptions about a model's reasoning process, undermining the reliability of MI studies (Elton, 2022). Madsen et al. (2024) discuss a concrete instance of this problem in probing research. Early BERTology work (Rogers et al., 2020) took high probing accuracy as evidence that neural representations encode linguistic properties directly. However, Hewitt and Liang (2019) showed that probes trained on *control tasks* – where labels are assigned arbitrarily rather than reflecting any real linguistic property – can achieve comparably high accuracy. This indicates that a probe's success partly reflects its

Table 4 Key challenges in mechanistic interpretability, with potential solutions and future directions

Challenge	Description	Potential solutions/future directions	References
Superposition	Individual neurons often encode multiple unrelated concepts, obscuring isolation of meaningful features	Sparse autoencoders and disentanglement methods for recovery of monosemantic features from superposed representations (Sect. 4.3)	(Elhage et al., 2022; Scherlis et al., 2025)
Spurious explanations	MI can uncover mechanisms that rely on coincidental or non-causal patterns in the data, potentially leading to misleading interpretations	Causal validation techniques (e.g., causal scrubbing) and standardized benchmarks (e.g., MIB, Interp-Bench) with known ground truth	(Elton, 2022; Méloux et al., 2025)
Scalability & deployment	MI is limited to small or simplified models; applying them to real-world problems remains technically infeasible and labor-intensive	Approximations of existing techniques; sparse circuit representations to reduce analysis scope; direct demonstrations of techniques at production scale	(Cammarata et al., 2020; Friedman et al., 2024)
Evaluation problems	Absence of universally accepted metrics for evaluating MI methods in terms of the faithfulness, completeness, or usefulness of interpretations	Benchmarks with shared tasks and metrics; development of standardized evaluation protocols enabling direct comparison across method families	(Hanna et al., 2024; Méloux et al., 2025)
Automation deficit	Circuit discovery methods require extensive manual work in both circuit analysis and intervention design, limiting reproducibility and scalability	Automated circuit discovery and feature extraction methods; LLM-assisted interpretation and automated hypothesis search to reduce case-by-case manual intervention design	(Conmy et al., 2023)
Semantic drift	The meanings of internal representations can shift across network layers or training checkpoints, challenging longitudinal analyses	Layer-wise probing methods such as lenses (Sect. 4.1) for tracking how representations evolve across depth	(Olah et al., 2020; Elhage et al., 2021)
Intervention pitfalls	Interventions may potentially influence the model behavior, making the causal importance of components and robustness of findings uncertain	Careful baseline selection; best-practice guidelines for patching to minimize unintended distributional shift	(Wang & Veitch, 2024)

own capacity to fit arbitrary patterns, rather than demonstrating that the underlying representation genuinely encodes the claimed structure.

Sharkey et al. (2025) and Räuker et al. (2023) emphasize the importance of validating hypotheses about models' behavior and the development of interpretability benchmarks. The comparison and validation of MI pursuits is only possible through the utilization of standardized evaluation techniques. Faithful interpretations should capture the true causal mechanisms driving model behavior rather than providing post hoc justifications that appear convincing but lack real explanatory power. Recent work by Méloux et al. (2025) addresses the identifiability of MI explanations, assessing both the uniqueness of discovered circuits and the algorithms they are claimed to implement. Their experiments reveal non-identifiability across every level of analysis: multiple distinct circuits can each reproduce the same model behavior; a single circuit can be assigned multiple competing interpretations; several different algorithms can be equally well causally aligned with the network; and a single algorithm can be causally aligned with different subspaces of the network. Together, these findings challenge the assumption that models implement unique, well-defined circuits for a

given behavior, underscoring the need for explicit identifiability criteria alongside any claim of having found "the" circuit responsible for a task.

This identifiability problem reinforces the broader need, discussed in Sect. 2.2, for standardized benchmarks (e.g., MIB Mueller et al. 2025b) and rigorous causal validation methods such as causal scrubbing (Chan et al., 2022). These tools should be employed to explicitly test whether a proposed explanation is sufficient to reproduce model behavior under controlled interventions, rather than relying on a single, potentially cherry-picked, demonstration.

Practical deployment. Mechanistic interpretability has seen only a limited adoption in non-research contexts. The following factors have contributed to this narrow utilization: the gradual emergence of MI into the mainstream (Saphra & Wiegrefe, 2024), MI research being limited to a family of models (Sharkey et al., 2025), and the continuous need for human input and validation (Bereska & Gavves, 2024). Current applications remain either experimental demonstrations or internal auditing tools, for example, the Golden Bridge Claude (Templeton et al., 2024) and other activation steering examples (Stoechr et al., 2024; Häon et al., 2025).

Moreover, results obtained on toy models may not be directly applicable to large-scale neural networks in the real world. Despite the advancement in the automation of MI workflows and the scaling up of techniques, further research is required to fully explore the potential of this approach to interpretability.

Universality. Interpretability is a difficult and laborious process that requires careful analysis from a human specialist. Each individual case needs independent interpretation of the results. Therefore, knowledge transfer between models and architectures could greatly accelerate advances in mechanistic interpretability. If the universality hypothesis (Chughtai et al., 2023; Merullo et al., 2024) is proven to be true, similar structures could be identified in multiple models. This would allow researchers to apply insights from one study to another.

Evidence for universality so far comes primarily from constrained settings. For instance, Chughtai et al. (2023) study universality of circuits emerging in one layer MLP architecture trained on different subsets of a group composition task dataset. Their findings show that all models implement the same algorithm, but using different representations. On the other hand, Merullo et al. (2024) further study the circuit discovered in Wang et al. (2022) and show that it implements simultaneously a different task: Colored Objects. Their findings suggest a different axis of universality – single circuit across tasks. However, the hypothesis has only been proven in toy models and for specific tasks. There is no guarantee that the results will be applicable to larger and more advanced architectures. Friedman et al. (2024) present the limitations of simplified model when met with out-of-distribution inputs. This suggests that even within a single model family, structural universality cannot be assumed without re-verification at each scale, let alone across architecturally distinct models. Work in this field remains an open challenge in MI, and is closely tied to the practical deployment challenges discussed above: without stronger evidence of universality, insights from research on smaller or toy models cannot be safely assumed to generalize to large-scale models.

Internal understanding. Mechanistic interpretability requires researchers to navigate the intricate landscape of the internal components and representations of a model. However, the task of understanding and translating these inner workings into human-understandable

processes is complicated by challenges such as superposition, semantic drift, and the effects of experimental interventions. Superposition entangles multiple features within the same neurons or activations, requiring careful disentanglement to isolate and interpret meaningful representations (Sect. 4.3). Individual neurons in small models have been shown to respond to several unrelated concepts simultaneously (Elhage et al., 2022; Scherlis et al., 2025), rather than encoding one clean, interpretable feature.

Semantic drift introduces additional uncertainty, as the meaning of features and concepts can shift across different layers of the network (Sect. 4.1, in particular lenses). For example, applying conditional probing (Hewitt et al., 2021), Hewitt et al. find that properties such as part-of-speech become accessible at deeper layers than prior, baseline-relative probing methods had suggested – illustrating how methodological choices in probing can themselves shape conclusions about where in a network a given property "emerges."

Moreover, the very act of intervening (Sect. 4.2) can itself alter the system's behavior, potentially confounding the interpretation. For instance, ablating or patching an activation can push the model into an out-of-distribution state not encountered during training, producing effects that reflect the model's behavior under an unnatural perturbation rather than revealing how the original, unperturbed computation actually worked (Zhang & Nanda, 2024).

Combined with cherry-picking and confirmation bias, these complexities lead to an incomplete or overly optimistic picture of a model's interpretability (Burns et al., 2023). As a result, ensuring rigor, transparency, and reproducibility is essential for avoiding biased conclusions in MI research. Recognizing and addressing these methodological pitfalls is critical to advancing a more reliable and scientific understanding of machine learning systems.

LLM focus. With the shift of mechanistic interpretability to transformers and language models (Saphra & Wiegrefe, 2024), relatively few researchers continue to dissect and interpret other model families. Transformer models are particularly attractive to MI researchers, as their residual stream allows for straightforward activation steering and embedding projections at any given stage. However, other model architectures remain widely deployed in real-world scenarios and require the same level of research attention.

For example, Hamblin et al. (2023) introduce circuit pruning, a saliency-based method that decomposes a CNN into sparse subnetworks, each responsible for computing a targeted latent feature – isolating, in some cases, sparser "subcircuits" tied to a feature's activation on particular inputs. This work directly demonstrates the viability of circuit discovery in CNNs, identifying feature-specific computational substructure analogous to the circuits sought in transformer-based MI research. Beaglehole et al. (2023) provide a rigorous mathematical background of feature learning, offering a theoretical grounding for how features emerge during training that could inform future CNN-specific circuit analysis – a level of theoretical rigor most empirical MI work currently lacks.

CNN-specific challenges – polysemantic channels, spatially distributed representations, and the absence of attention-like routing mechanisms – are left comparatively understudied. To the best of our knowledge, there is no standardized benchmark for circuit study in CNNs, echoing the broader benchmarking gap noted in the *Evaluation problems* challenge above.

5.2 Opportunities

Despite the multiple challenges present in MI, it is still a promising area of research. The possibilities in the context of AI safety convince researchers to continue work in this field. We outline the key opportunities, followed by a detailed description.

Potential applications and benefits of applying MI are:

- *Debugging*: MI enables the identification of faulty reasoning or unexpected model behavior by tracing decisions to specific circuits or internal mechanisms, helping to fix or retrain models more effectively (Olah et al., 2020; Wang et al., 2022).
- *Data privacy*: By understanding how models store and access information internally, MI can help reveal whether sensitive training data is memorized or indirectly encoded, aiding privacy audits (Carlini et al., 2023).
- *Robustness*: MI can highlight the features and mechanisms that models rely on, exposing vulnerabilities to adversarial inputs or distribution shifts and guiding strategies to make models more robust (Burns et al., 2023; Ilyas et al., 2019).
- *Control*: Gaining mechanistic insight opens the possibility of directly editing model behavior by modifying or steering internal components, supporting efforts in fine-tuning and alignment (Sharkey et al., 2025; Meng et al., 2023).
- *Human-in-the-loop*: MI supports collaborative AI development by allowing humans to interpret and interact with models' decision processes, enabling feedback loops for supervision and correction (Chung, 2021).
- *Trustworthiness*: By revealing the internal logic behind model outputs, MI can provide evidence of consistent, rational behavior, supporting confidence in AI systems used in high-stakes domains.

AI safety. Mechanistic interpretability is focused on establishing the roles of components in neural networks – neurons, attention heads, or whole layers. MI not only relies on the input–output relations, but also studies the causal dependencies in between, thus dissecting the model's decision-making process step by step. This makes MI methods reliable tools for locating features – both valuable, such as factual information (Meng et al., 2022; Geva et al., 2023; Fierro et al., 2025) or truth representations (Marks et al., 2024), and dangerous, such as hallucinations (García-Carrasco et al., 2024; Li et al., 2024; Yu et al., 2024) or toxic neurons (Yang et al., 2024). Li et al. (2025) further extend this idea by locating and removing redundant parameters, demonstrating that MI insights can be applied to design more robust and efficient models.

Mechanistic interpretability seeks to elucidate the inner workings of neural networks in order to increase their transparency and trustworthiness. MI has been applied in the task of knowledge localization, debugging, and removal of harmful components.

Algorithmic understanding of neural networks. Mechanistic interpretability aims to identify *how* information is processed within neural networks. This involves reverse-engineering internal circuits and structures, and expressing them in human-interpretable terms. MI offers a more granular understanding of a model's decision-making processes by recovering computational routines that resemble hand-engineered algorithms. Achieving algorithmic understanding holds significant promise for improving our ability to reason about model generalization, identify failure modes, and precisely intervene on internal behavior.

For instance, Chen et al. (2024) identified a circuit responsible for computing document relevance in the context of document ranking.

Real-world applications. While much of the foundational work in MI has focused on general-purpose AI systems, recent studies have explored its application in specific domains. We highlight finance and medicine as illustrative domains, as both are high-stakes, heavily regulated fields where the need for transparent, auditable AI decision-making is particularly acute, making them natural test cases for the practical applicability of MI – and, in the case of medicine, for illustrating the current gap between this need and existing work. For instance, in finance, Golgoon et al. (2024) applied mechanistic interpretability techniques to GPT-2 Small, aiming to uncover how the model identifies potential violations of Fair Lending laws. To the best of our knowledge, no comparable application of mechanistic interpretability has yet been reported in medicine. Nevertheless, prior work has emphasized the necessity of transparent and interpretable AI systems in clinical settings (Band et al., 2023; Wang et al., 2024), suggesting that MI – with its emphasis on causal, mechanism-level explanations rather than correlational ones – could be particularly well-suited to domains where understanding *why* a model reached a decision is as important as the decision itself.

6 Conclusions

Mechanistic interpretability has emerged as a promising research program within the broader landscape of explainable artificial intelligence. In recent years, deep learning has achieved substantial progress across diverse domains, including natural language processing, computer vision, and scientific discovery. Despite these advances, a fundamental challenge remains: understanding the mechanisms by which these models generate their predictions. MI addresses this issue by aiming to uncover the underlying computations within neural networks, providing insight into the specific mechanisms that give rise to observed behaviors. This survey sought to provide a comprehensive entry point into the field of MI. We introduced MI by providing its historical and scientific context (Sect. 2). We presented a unified taxonomy of MI approaches (Sect. 2) and explored in detail multiple techniques, provided examples and pseudo-code for representative MI methods (Sect. 4). Our literature review demonstrated the rapidly growing interest in the field (Sect. 3).

One of the central promises of mechanistic interpretability lies in its potential to move beyond post-hoc correlations and surface-level attributions – common in other XAI approaches – toward causal, structural, and component-level understanding of neural network behavior. MI aspires to uncover algorithmic structures that are implemented within the weights and activations of trained models. By reverse-engineering these internal mechanisms, researchers can gain insight into the representations learned by the model, the decision-making procedures it implements, and the ways in which it generalizes (or fails to generalize) across tasks and inputs.

A strong mechanistic understanding is advantageous in a number of ways. MI could improve model debugging and failure mode analysis by identifying the specific pathways that contribute to undesirable behaviors. It may enable model editing – the ability to intervene on internal components to change a model's outputs in a controlled and interpretable way. An important direction for future research is the transfer of circuits between models, addressing whether and how discovered functional circuits from one model can be used in

another. An additional area worth exploring is the representation and integration of different modalities in mechanistic circuits. Furthermore, MI holds the promise of supporting safety and alignment efforts, particularly in the context of powerful foundation models, where understanding the internal dynamics of goal representation, deception, or power-seeking behavior may be critical. MI could also contribute to the development of more scientifically grounded understanding of AI, with machine learning systems serving not only as effective tools but also as objects of study in their own right – similar to the manner in which the study of natural intelligence is conducted in neuroscience and cognitive science.

At the same time, the field of MI is still in its early stages and many challenges remain. Among these are scaling interpretability techniques to the complexity of frontier models, developing principled evaluation frameworks for mechanistic claims, and determining the limits of human comprehensibility when confronted with distributed representations and high-dimensional feature spaces. Open questions also remain regarding the degree to which mechanistic insights transfer across architectures, training regimes, or tasks.

We view this article as a foundational step toward consolidating the field of mechanistic interpretability. By providing a coherent definition, a taxonomy of techniques, and a structured overview of the current research landscape, we hope to support further theoretical development, methodological refinement, and interdisciplinary dialogue. The long-term vision of MI is ambitious: to build a science of deep learning systems that enables not only prediction and control but also understanding.

Author Contributions B.K. conceived the project and wrote the manuscript. H.K. supervised the work and contributed to the literature research. Both authors reviewed and approved the final version of the manuscript.

Data Availability Bibliographic data was retrieved from Scopus and Web of Science. Access to these databases is subject to institutional or individual licenses and is not publicly available. The data were used in accordance with the terms and conditions of the respective providers.

Declarations

Conflict of interest The authors declare no conflict of interest.

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